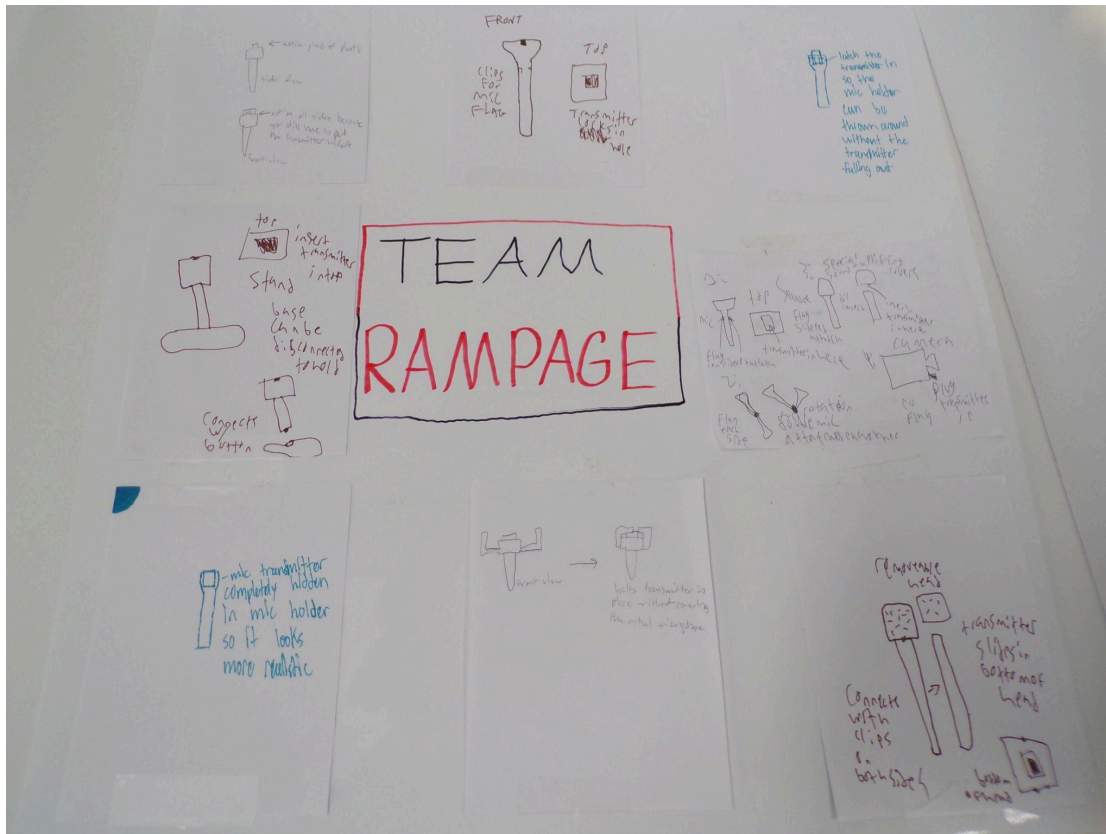


Element D: Design Concept Generation, Analysis, and Selection

Brainstorming Session: 2/6/24



Collaboration: 2/8/24

Design Idea Generation (D1)

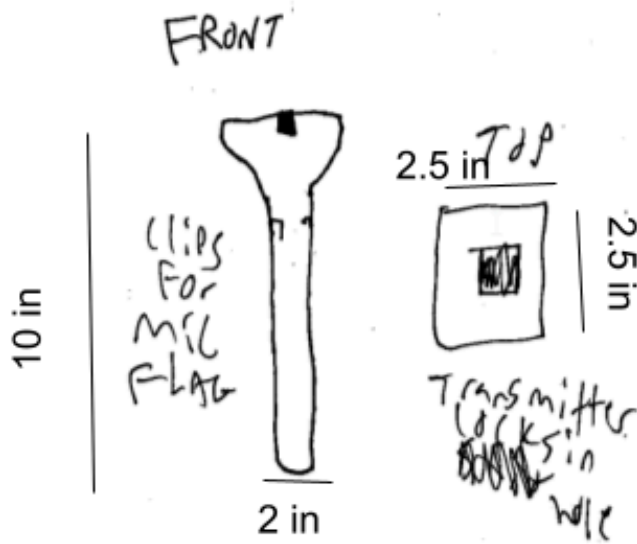
Design Idea #1 Microphone holder with slot for transmitter on top

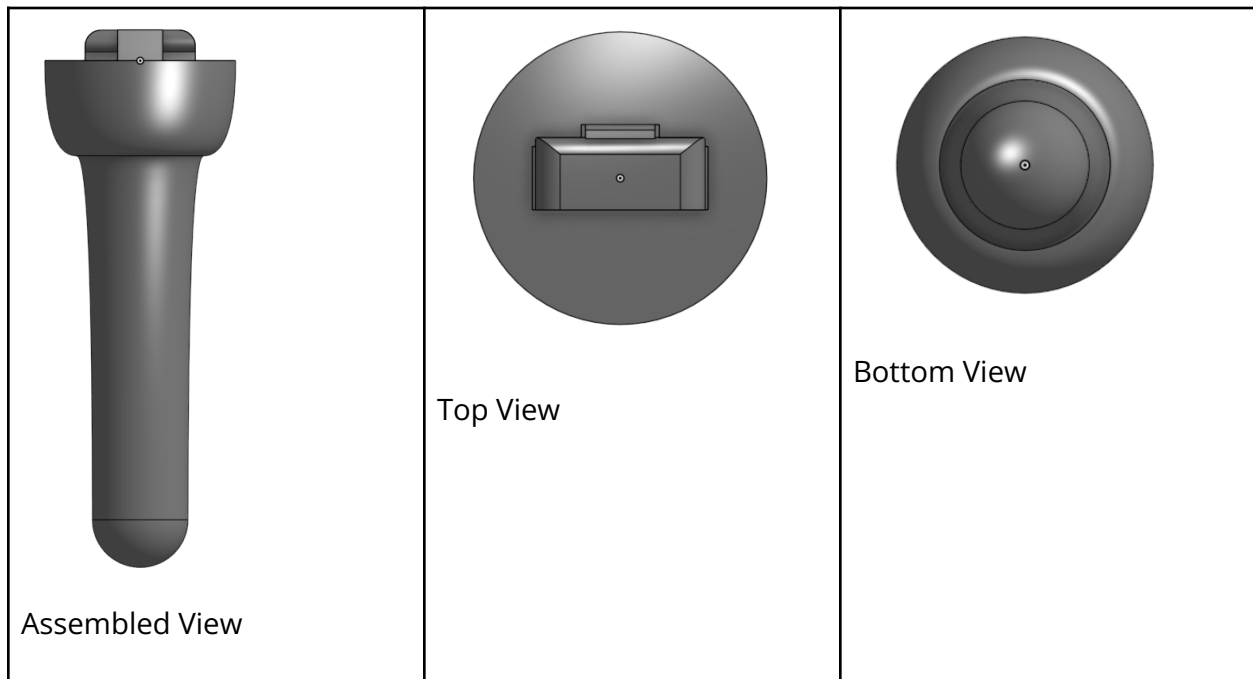
Date: 2/6/24

Description: Standard microphone design, with handle extending into larger diameter, microphone transmitter goes in the top, in a slot that will be long enough to cover the entire transmitter.

Drawing that includes distinguishing information such as scale, dimensions, and materials:







Feedback: Similar to previous iterations which all satisfied some of our design criteria. Not certain about the shape of the top of the holder, since the transmitter is a rectangle. Not yet sure how the mic flag clips will deploy, although the idea is promising.

Design Idea #2 Holder with removable head to insert and hide mic transmitter

Date: 2/6/24

Description: Head of mic holder is able to be removed from holder shaft. The transmitter is put inside the head cover so it is completely hidden.

Drawing that includes distinguishing information such as scale, dimensions, and materials:

Feedback: Promising idea of how the transmitter will sit completely enclosed in the microphone holder, but not sure about the attachment mechanism between the top and bottom parts of the holder. Also, since the transmitter has a secure clip on only the bottom side of it, a new solution will have to be developed.

Date: 2/28/24

https://www.youtube.com/shorts/q_vryt-MXz4

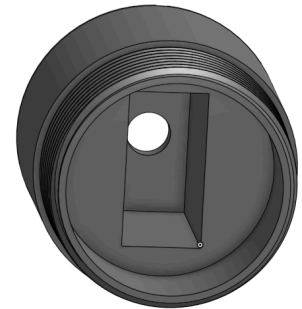
Iteration 1.0



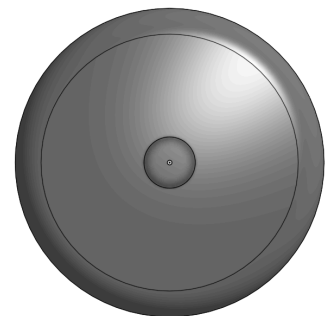
Exploded View



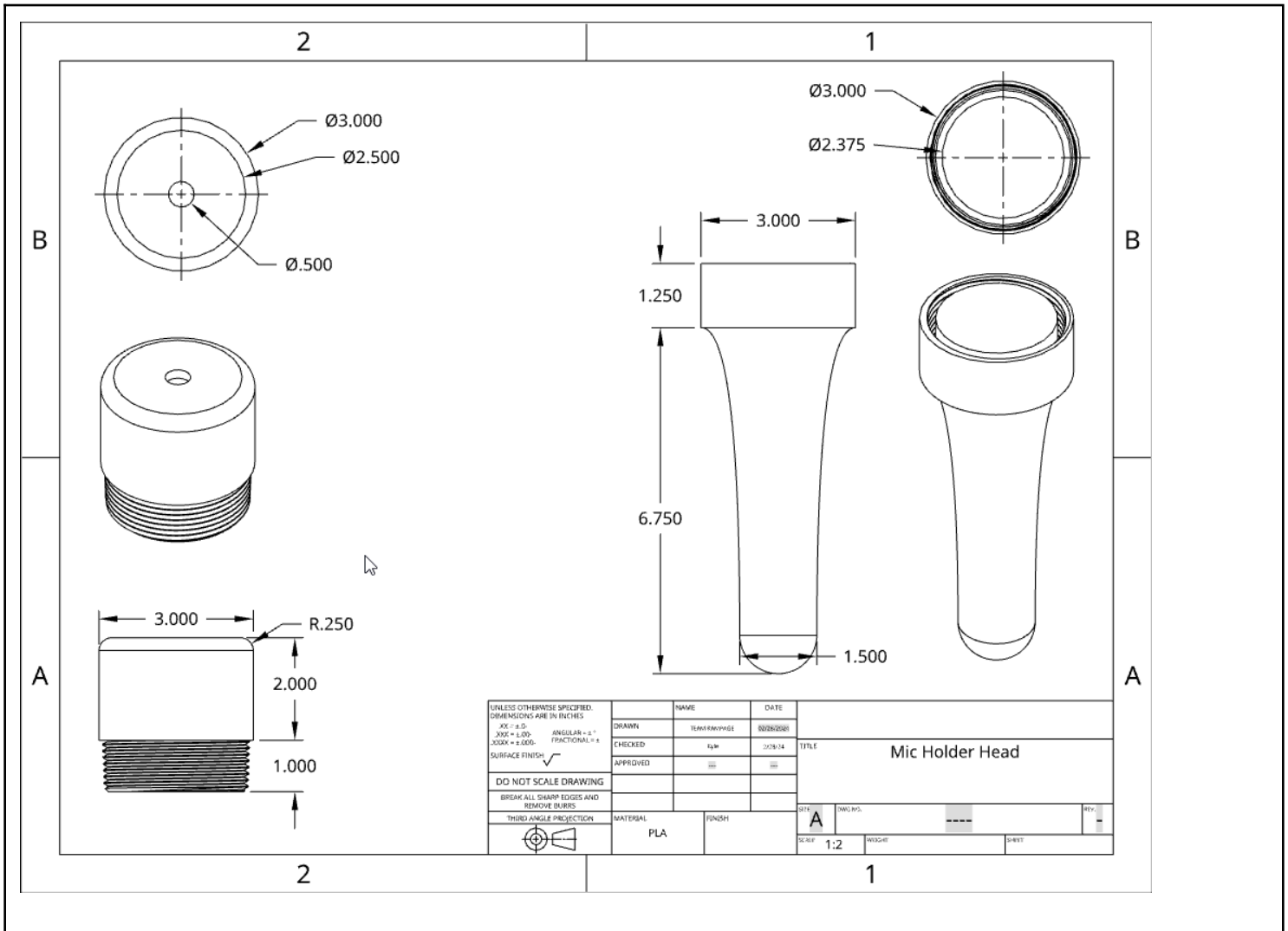
Assembled view



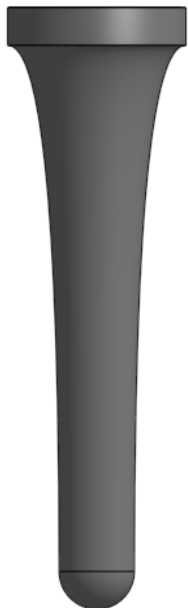
Opening view



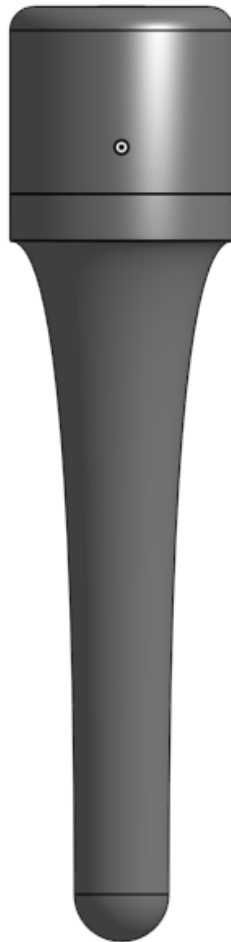
Top View



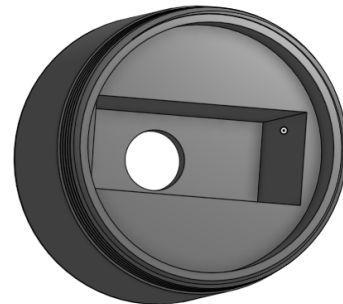
Iteration 2.0



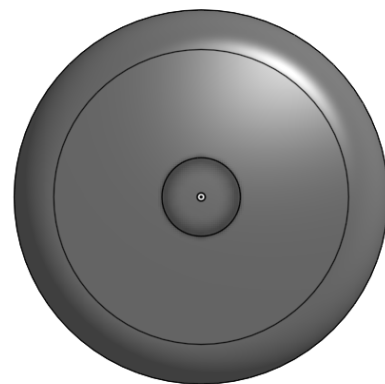
Exploded View



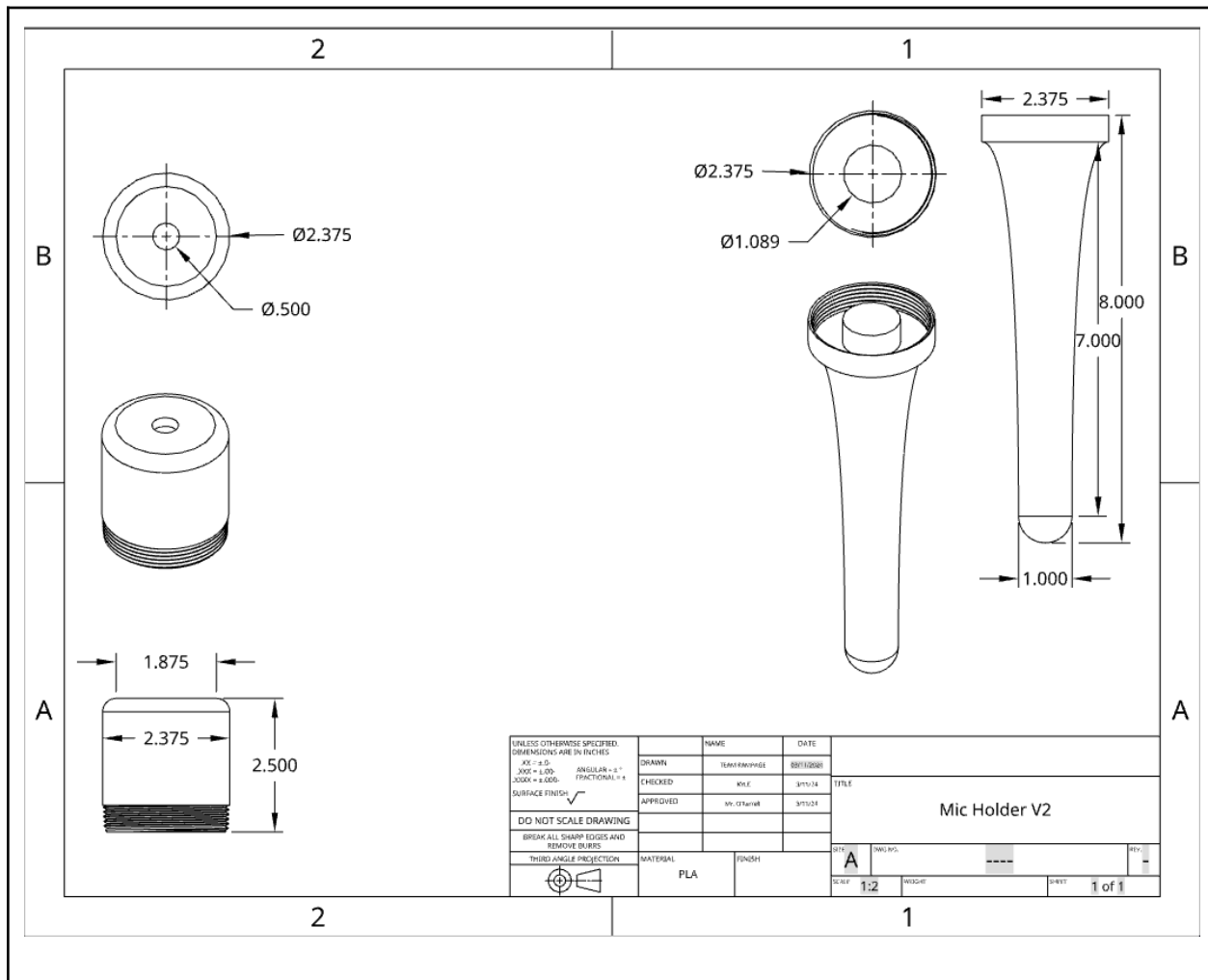
Assembled View



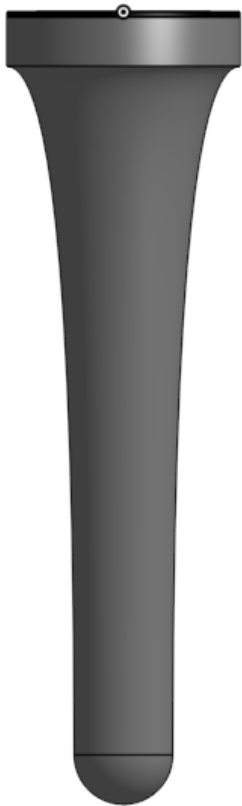
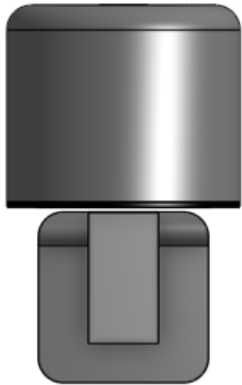
Opening view



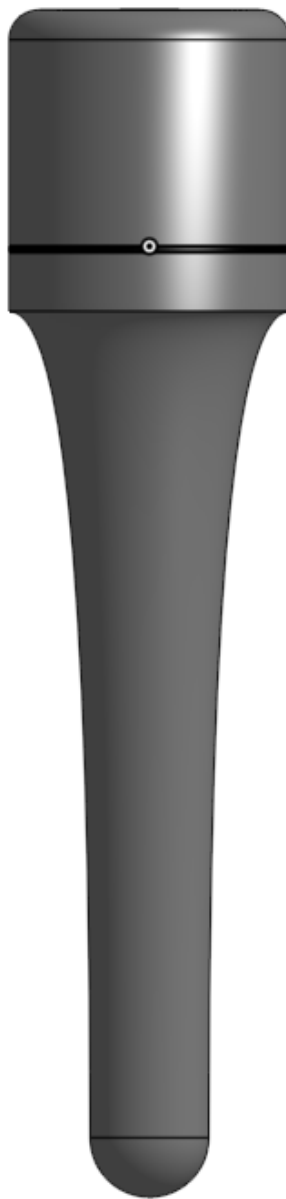
Top View



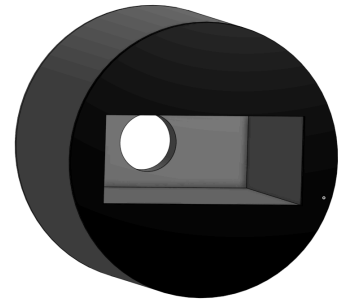
Iteration 3.0



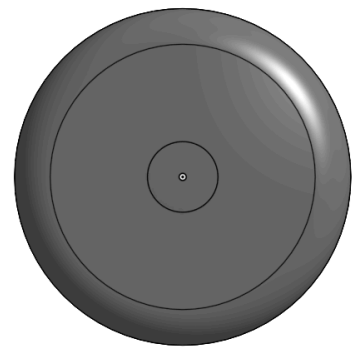
Exploded View



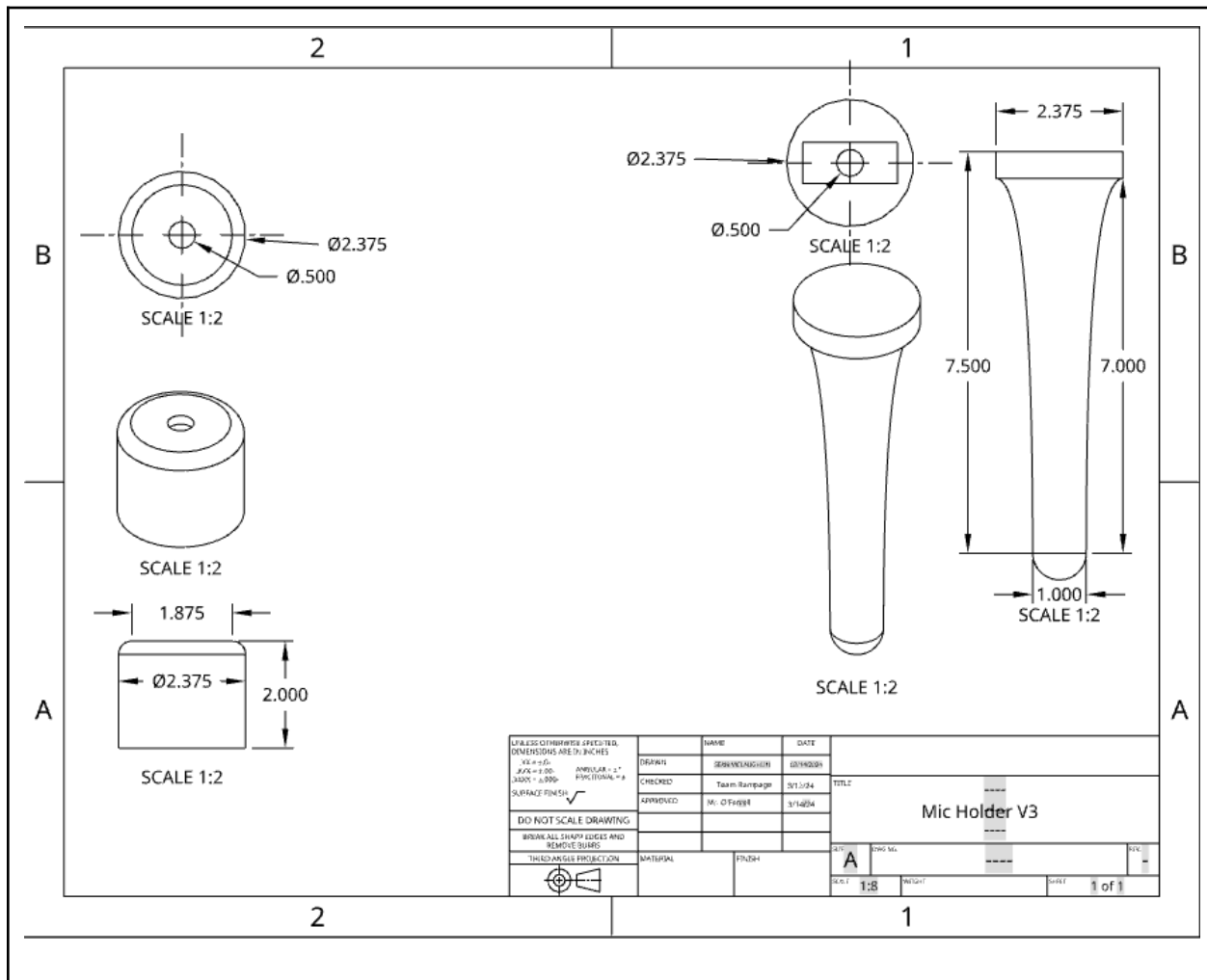
Assembled View



Opening view

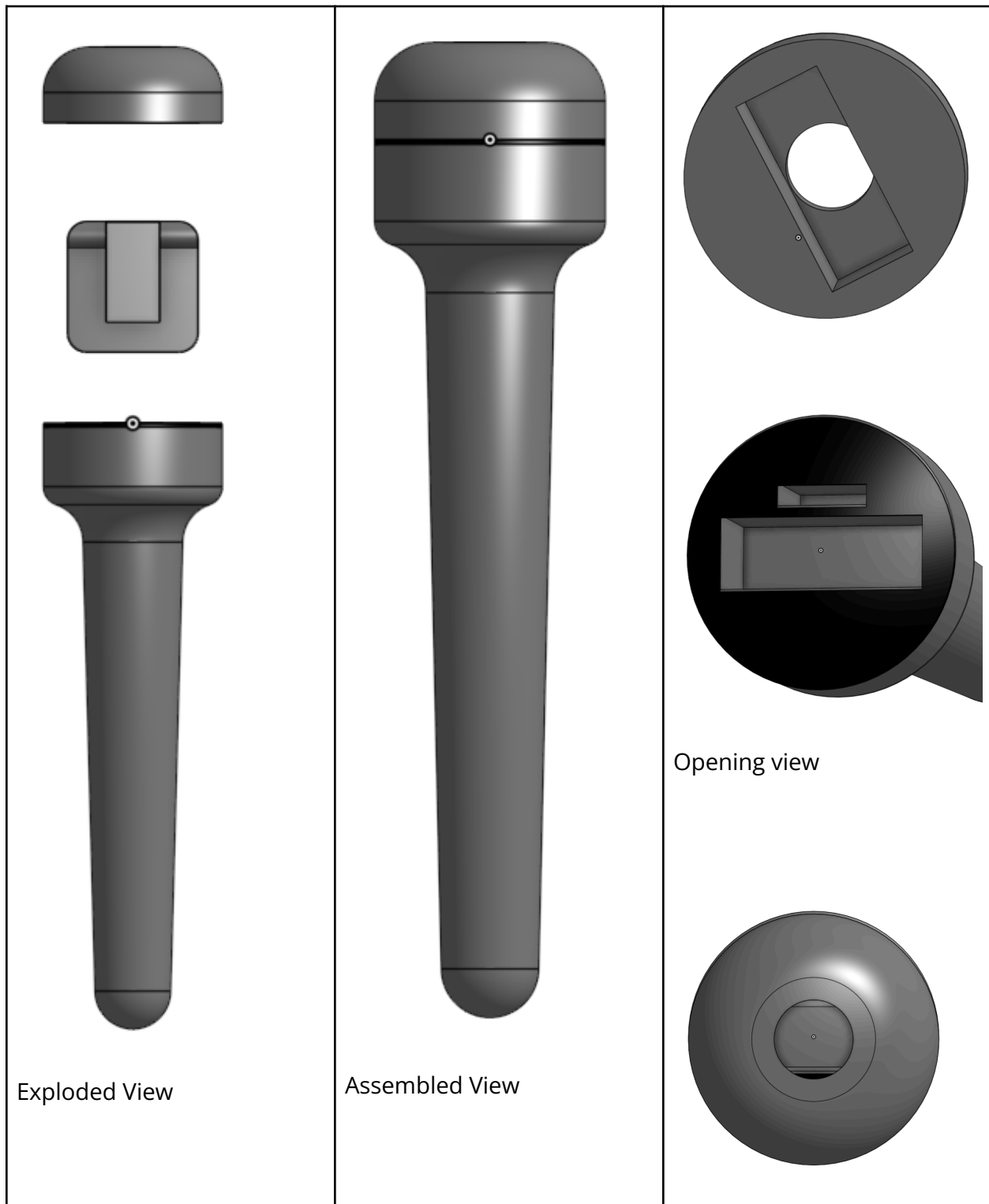


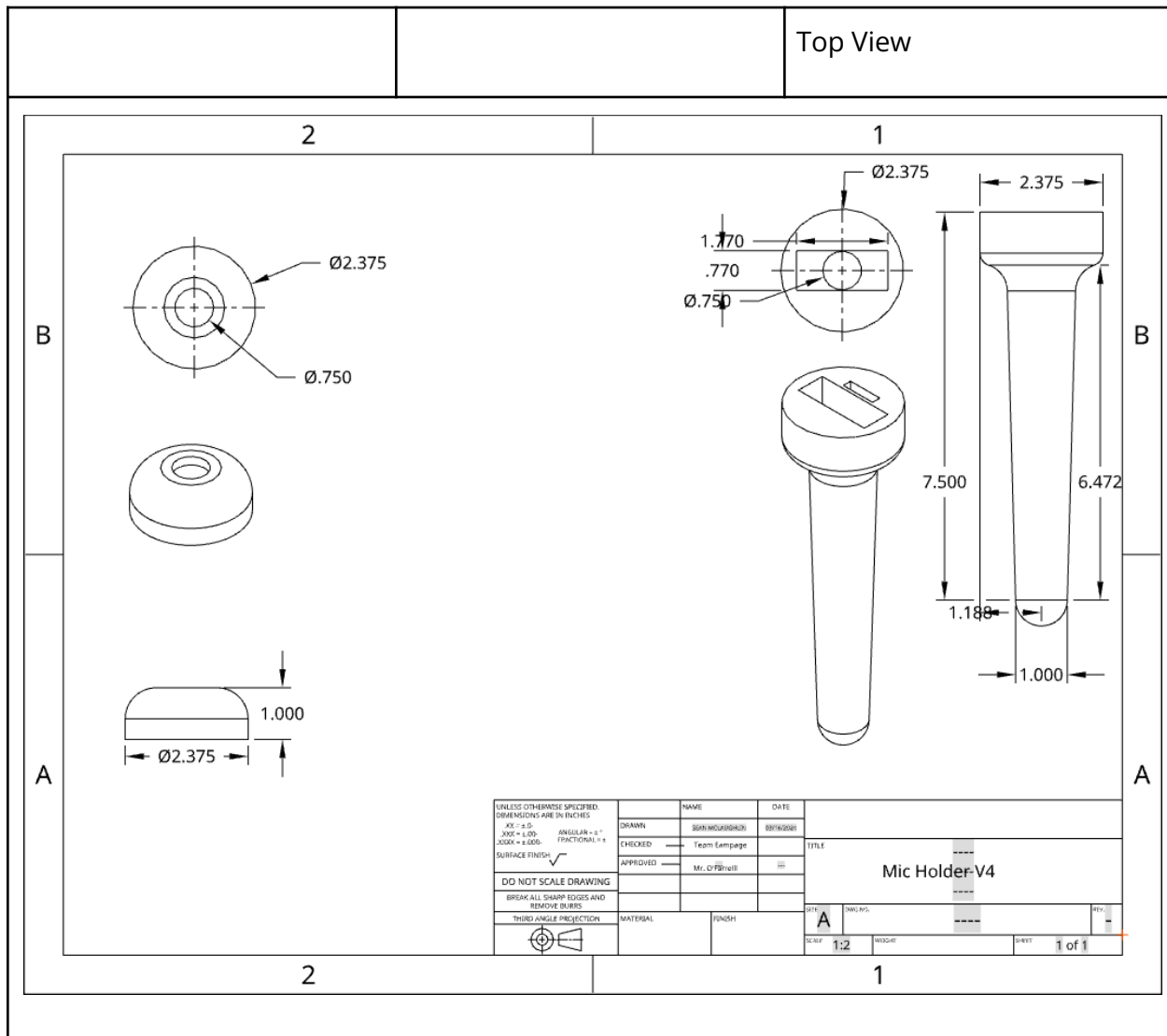
Top View



Iteration 4.0







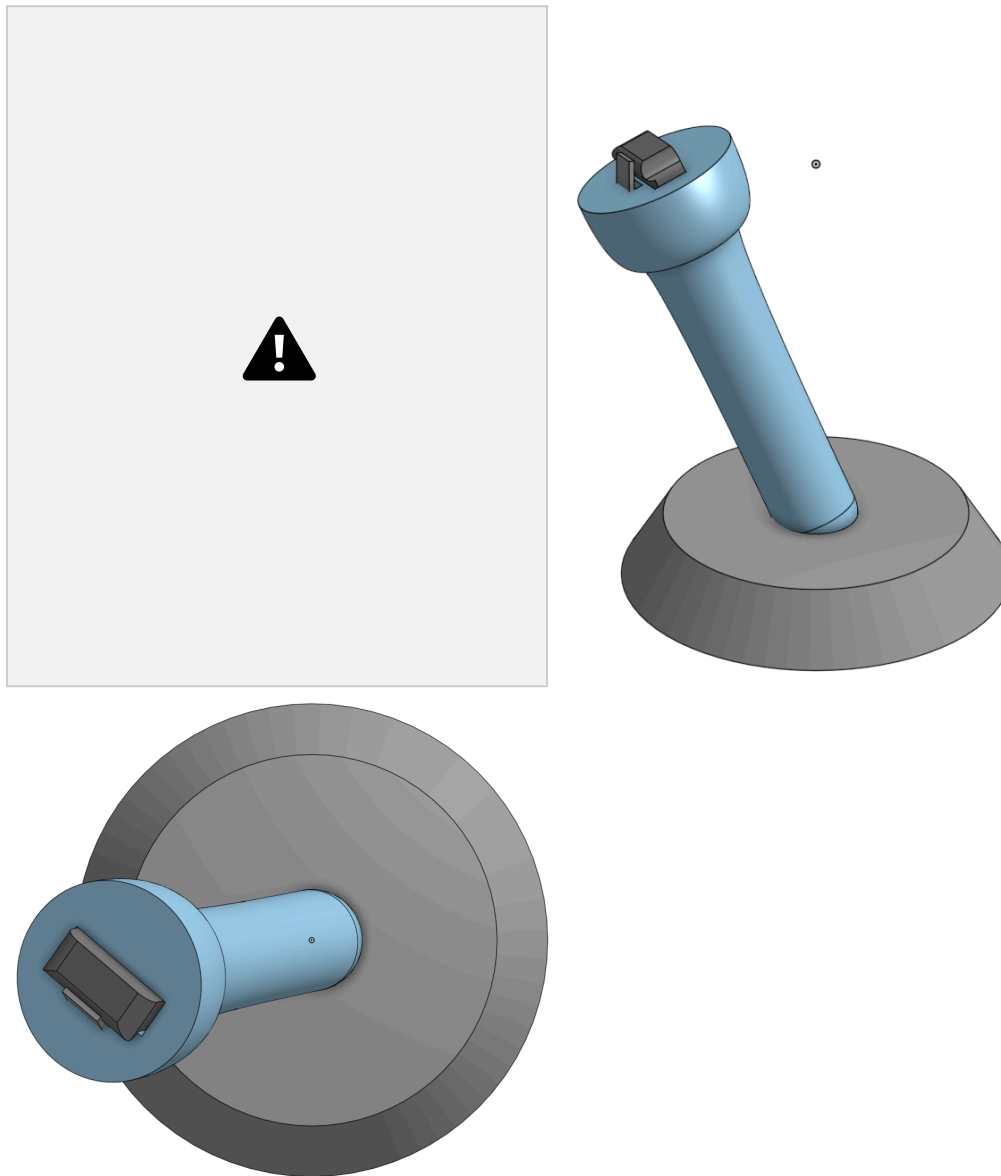
Design Idea #3 Holder with slot in top for insertable mic transmitter; detachable stand

Date: 2/6/24

Description: Same as first design, but a bit smaller and with a removable base. The base allows the microphone to stand on a surface, such as a desk. It can be removed to interview, but it would be smaller than the first design.

Drawing that includes distinguishing information such as scale, dimensions, and materials:





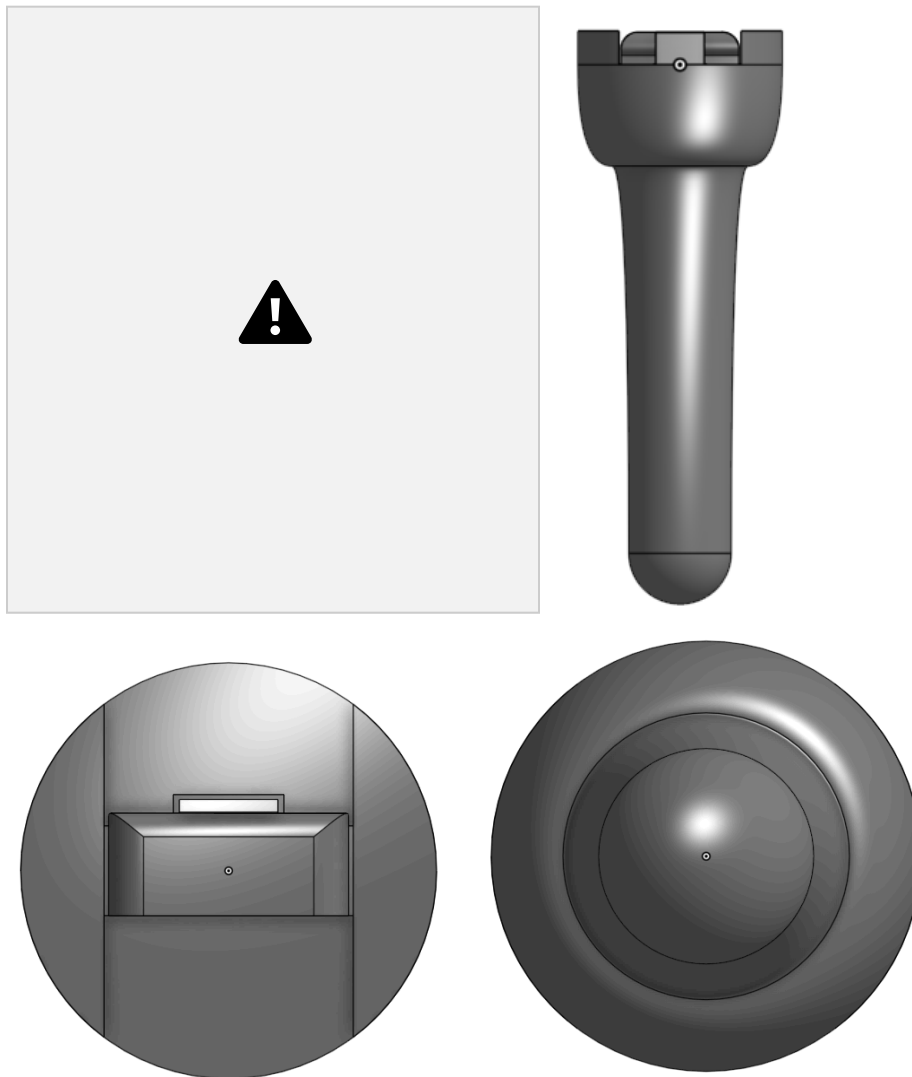
Feedback: Similar to previous iterations which all met some of the design criteria. Stand could be useful at times, but ultimately it would prove inconvenient given the microphones will be taken around the school instead of being laid on a table.

Design Idea #4 Mic hold with two sided transmitter cover

Date: 2/8/24

Description: This design draws comparisons to older designs which met some of our design criteria. It holds the transmitter in place using the transmitter's clip, and adds a sidewall on the left and right sides to reduce mobility for the transmitter after it is locked in.

Drawing that includes distinguishing information such as scale, dimensions, and materials:



Feedback: I like the idea, it would definitely work to hold the mic transmitter in place. I think that having it on two sides as opposed to four would work well to give it the ability to be removed. I am wondering if the transmitter having a bit of give to it would create sound interruptions? If an interviewer swung it and the transmitter hit off of the sides or top

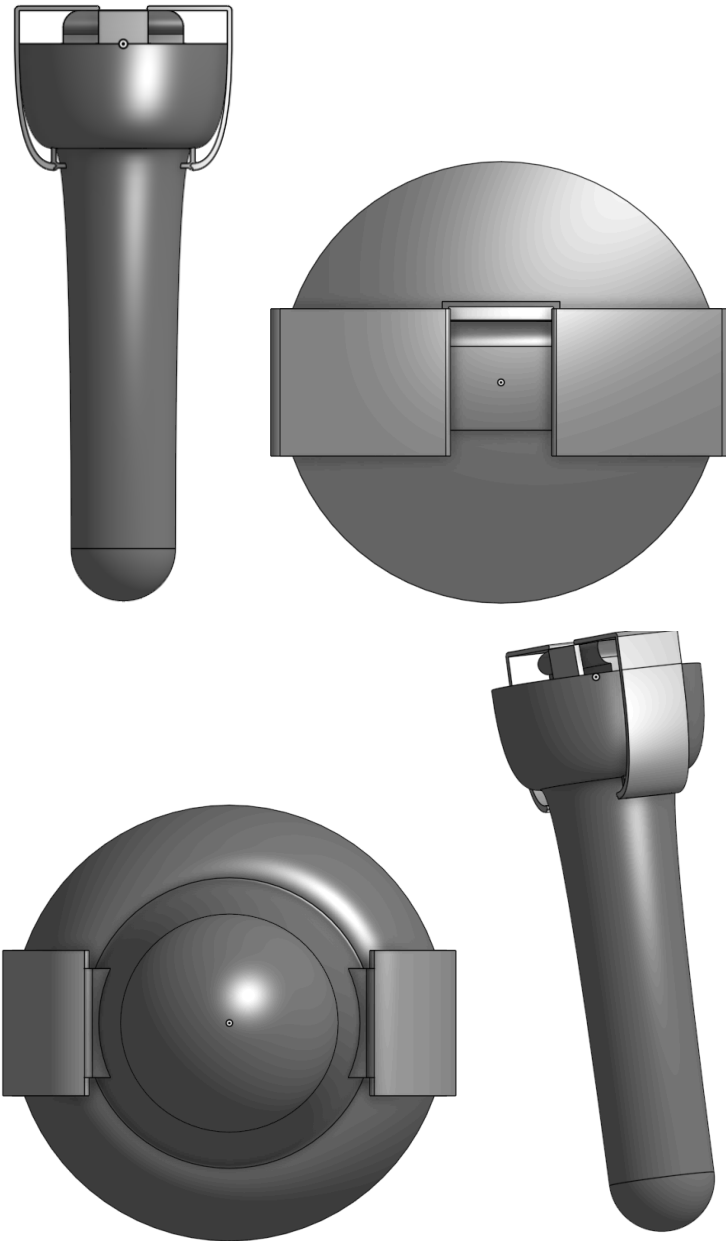
would it make a “thump” sound? Overall, I like the design and it would definitely work well. It would do its job perfectly and hold the transmitter.

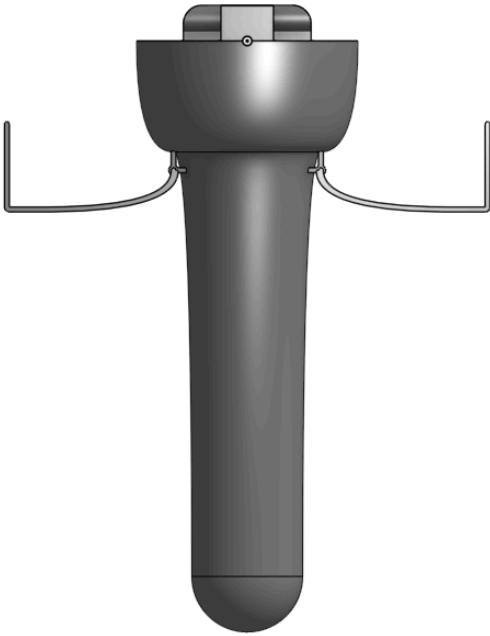
Design Idea #5 Holder with clamps to cover mic transmitter

Date: 2/8/24

Description: This design is also similar to previous iterations which met some of our criteria. One of our criteria was that the transmitter would stay in place when the holder is moved, so this design includes one wing on each side which flips upwards and locks the transmitter in place. They notably do not cover the actual microphone input on the transmitter, which is in the middle.







Feedback: This is a really good idea. I really like the concept of the hinges being able to come off or on. I like how it does not cover the top of the transmitter so that the sound is clear. The concept of using hinges would be good for a student to use because it is explicit that it can only move over top to cover the transmitter. It would do a perfect job of holding the transmitter. The hinges could be designed to look like the top of a microphone to give it a look of authenticity.



Design Idea Comparison and Selection (D1, D2)

	weight	Idea 1 - <i>Mic holder with slot for transmitter on top</i>	Idea 2- <i>Holder with removable head to insert and hide mic transmitter</i>	Idea 3 - <i>Holder with slot in top for insertable mic transmitter; detachable stand</i>	Idea 4- <i>Mic holder with two sided transmitter cover</i>	Idea 5- <i>Holder with clamps to cover mic transmitter</i>
Authenticity	30%	3	5	2	4	1
Security	30%	2	4	1	3	5
Comfortable	15%	5	5	4	5	5
Weight	15%	5	3	1	4	2
Size	10%	5	3	1	4	2
score		3.5	4.2	1.75	3.85	3.05
ranking		3	1	5	2	4

Ranking Scale of 1-5 -- 1 is best, 5 is worst

Plan of Action (D3)

A Gantt chart is encouraged to lay out the primary steps expected on a visual timeline. Larger tasks from the Gantt chart should be broken down further into detailed sub-tasks, which may be housed in a different day-to-day document. Discrete tasks should include enough detail such that others could replicate the efforts, such as procedures, materials, tools, status, etc.

	2/26	2/27	2/28	2/29	3/1
CAD Designs	Design mic transmitter	Design holder	Design head	Prepare to print (slicing software)	Start printing

	3/1	3/4	3/6	3/8	3/11
3D Printing Prototypes	Print first prototype	Revise first prototype	Print second prototype	Revise second prototype	Print third prototype

	3/11	3/12	3/13	3/14	3/15
CAD/3D Printing					

^Will be updated as the project goes

Design Blueprint (D4)

Design Blueprint #1

Date: 2/26/24

Sketch: First iteration

Manufacturing Methods:



- 1: 3D print
- 2: Construction out of cardboard
- 3: Paper modeling

Supply List and Costs:

Item	Cost each	# needed	Total
Polyterra PLA	\$1.77	5	\$8.85
Cardboard	\$5.92	5	\$29.6
Total Cost	\$7.69	5	\$38.45

***Our Prusa 3D Printer is provided by the school**

